Tech Careers

Emerging Dimensions

The sheer range of industries it finds a place in, and the increasing demand for trained professionals, marks CAD/CAM as a very attractive career option

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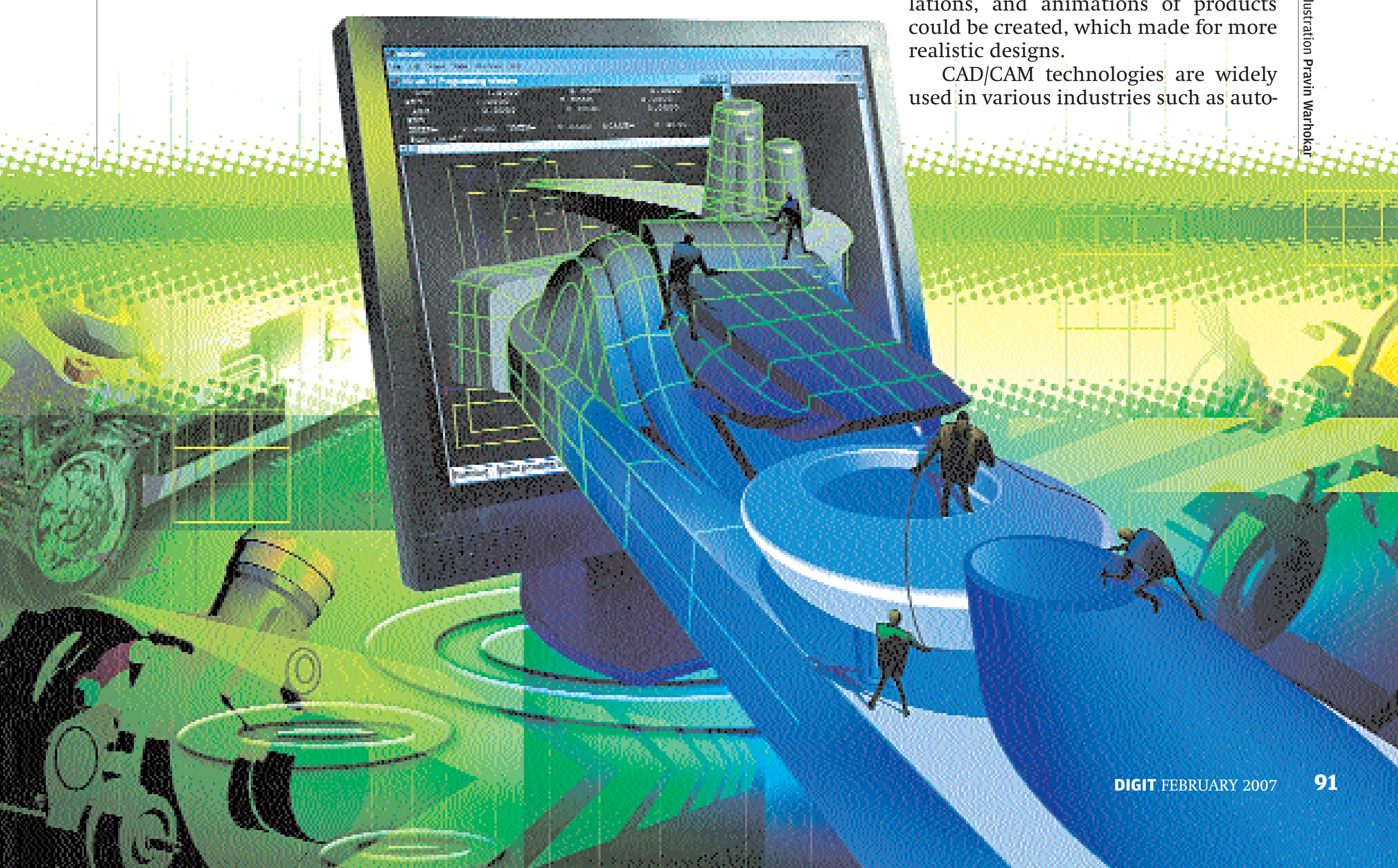
The Taj Mahal is an excellent example of architectural design. The design and construction of this marvel of a monument took years to complete—which is often quoted as testimony to its greatness. But we must note here, without disrespect to those architects long gone, that were they to have had access to Computer Aided Design (CAD) technology, their work would have been much easier, and execution would have progressed much faster.

CAD and its allied technologies—Computer Aided Manufacturing (CAM) and Computer Aided Engineering (CAE) has widened the horizons of organisations at every level. 10 or 15 years ago, designs of a product, vehicle, or building were drawn by hand using special geometrical drawing tools. It would take several weeks or even months for a drawing to get to completion.

Gradually, as the IT revolution hit every industry, developments took place in the field of CAD/CAM. ("CAD/CAM/CAE" is usually referred to as just "CAD/CAM", since CAE just involves tasks that involve analysis, testing, and diagnostics on products or assembly lines, for instance.) Computers began to be used for drafting. 2D modelling was initially developed to minimise the time consumed for drawing the design of a new product, optimising existing products, and deriving better results. When CAD/CAM technologies gradually shifted from 2D to 3D, advanced solutions for designing a product could be derived. Walkthroughs, simulations, and animations of products could be created, which made for more realistic designs.

CAD/CAM technologies are widely used in various industries such as auto-



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